

2025 Chipathon Padframe Integration Track

Prof Morbius

5V

3.3V core

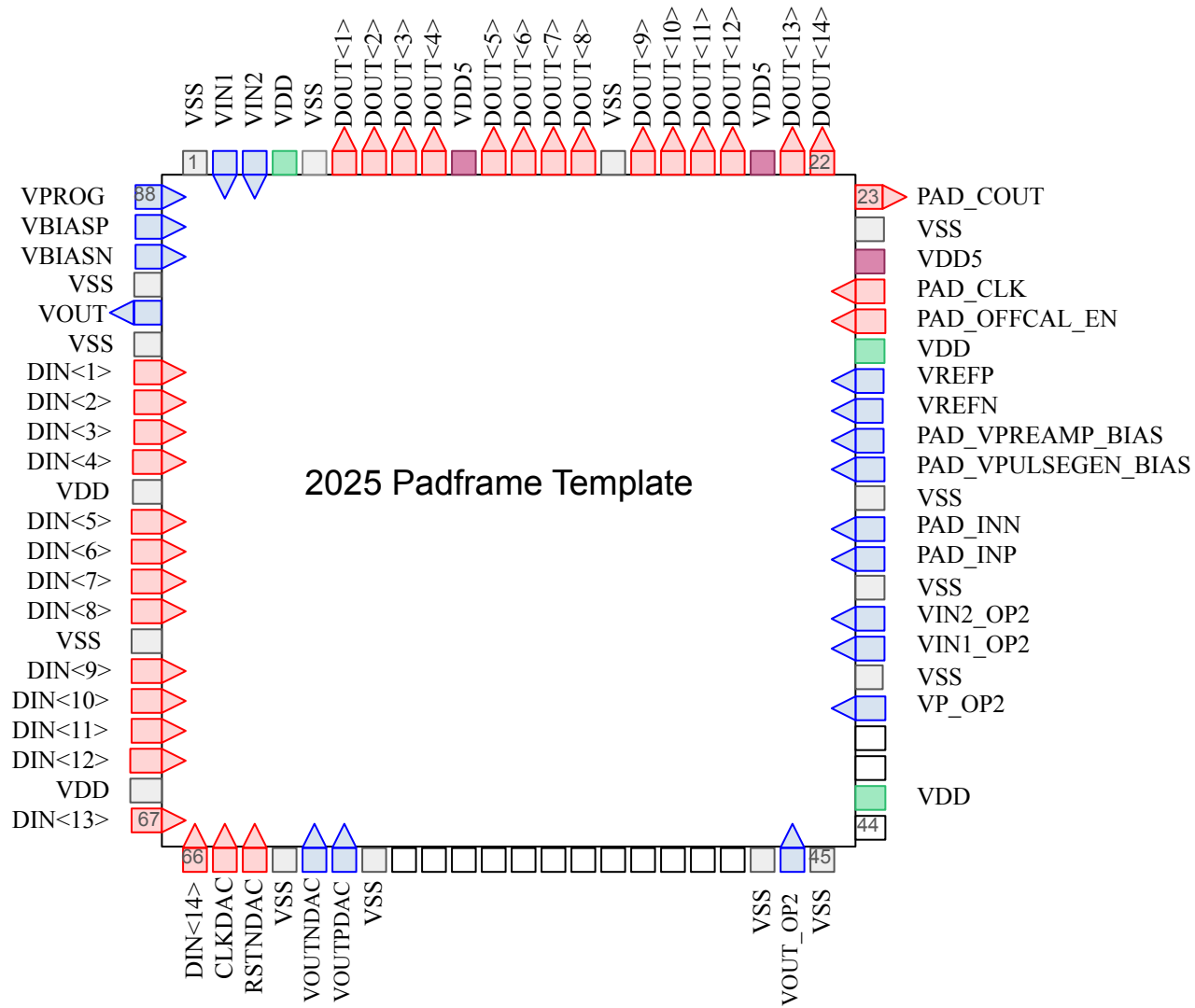
Analog signals

Digital signals

ground

N/A

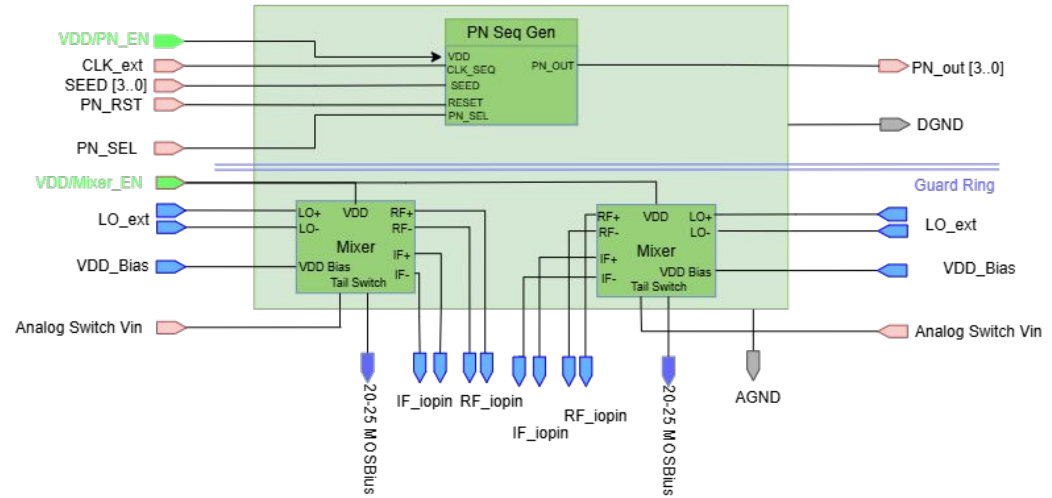
Space for 88 pins



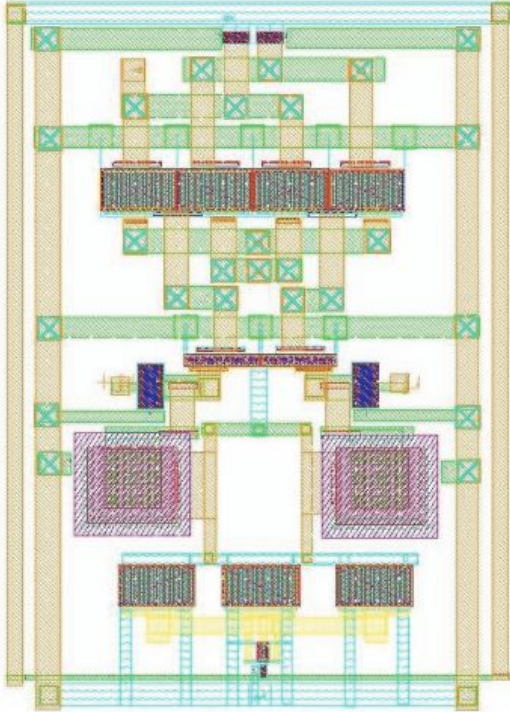
Total of pins used 33 pins

Types of available pins:

- 5V
- 3.3V core
- Analog signals
- Digital signals
- ground
- N/A



Total (Estimated Die)



Double balanced gilbert cell

Component	Estimated Area (180nm)
Gilbert Cell Mixer	~0.0084 mm ² (70μm x 120μm)
SPDT T/R Switch	0.019 mm ² (160μm x 120μm)
PN Sequence Gen.	~0.002 mm ² (40μm x 50μm)
Total Core Area + Free Space	~0.0336 mm ²

Total Die area including the guard ring + Pad Connection + ESD Ring protection = 0.64mm²

The core will have a total area of 0.0336mm²

PIN List

Number	Name	Type	Direction
1	VDD/Mixer_EN	5V Power	Input
2	DGND	Ground	Input
3	AGND	Ground	Input
4	RF+_R	Analog	Bidirectional
5	RF-_R	Analog	Bidirectional
6	LO_ext+_R	Analog	Input
7	LO_ext-_R	Analog	Input
8	IF+_R	Analog	Bidirectional
9	IF-_R	Analog	Bidirectional
10	VDD_Bias_R	Analog	Input
11	Analog Switch		
	Vin_R	Analog	Input
12	RF+_T	Analog	Bidirectional
13	RF-_T	Analog	Bidirectional
14	LO_ext+_T	Analog	Input
15	LO_ext-_T	Analog	Input
16	IF+_T	Analog	Bidirectional
17	IF-_T	Analog	Bidirectional

17	IF-_T	Analog	Bidirectional
18	VDD_Bias_T	Analog	Input
19	Analog Switch		
	Vin_T	Analog	Input
20	VDD/PN_EN	3V Power	Input
21	CLK_ext/PN_EN		
	N	Digital	Input
22	Seed_IN 0	Digital	Input
23	Seed_IN 1	Digital	Input
24	Seed_IN 2	Digital	Input
25	Seed_IN 3	Digital	Input
26	PN_RST	Digital	Input
27	PN_SEL	Digital	Input
28	PN_out 0	Digital	Output
29	PN_out 1	Digital	Output
30	PN_out 2	Digital	Output
31	PN_out 3	Digital	Output
32	MOSBius Current Mirror_R		
		Analog	Bidirectional
33	MOSBius Current Mirror_T		
		Analog	Bidirectional